AE 6355 Computer Project Cover Sheet

Base Project __________ pts out of 60

Project Report _________ pts out of 20

Add-ons, put an x on the line in front of the add-on if you completed it:

_____ 1) Developer defined: ________ pts out of _______

_____ 2) Non-planar equations of motion _______ pts out of 10

    _____ a. Monte-Carlo _____ pts out of 5

    _____ b. Arbitrary vehicle geometry ______ pts out of 5

__x___ 3) Optimize Deorbit for heating and/or deceleration _____ pts out of 10

__x___ 4) Calculate aero coefficients _______ pts out of 5

    _____ a. Stag point heat rate with Fay & Riddell _____ pts out of 5

    _____ b. Tauber-Sutton radiative heating ______ pts out of 3

    _____ c. Use CO_2 EquilFlow input for Mars/Venus ______ pts out of 2

_____ 5) Use TPS response surface equation ______ pts out of 10

__x___ 6) Arbitrary Atmopshere _______ pts out of 3

    __x___ a. J2 gravity model _____ pts out of 2

_____ 7) 1D finite difference solution of the heat equation _____ pts out of 20

_____ 8) Parachute and terminal descent _____ pts out of 20

Planetary Entry, Descent and Landing Final Project

Geoffrey Hendricks*
Georgia Institute of Technology, Atlanta, GA, 30332

This report discusses the development of a series of code used to simulate a generic Earth entry mission from any orbit using Stardust's vehicle geometry. With flexibility of input parameters like velocity at and flight path angle at atmospheric interface as well as atmospheric model selection for density, the code can be used to model a number of different entry trajectories and determine their feasibility. Furthermore, this report will discuss how to determine data such as the entry corridor, max heat rate, and total heat load which are necessary to define successful entry trajectories and appropriate heat shielding designs. Lastly, the code will be validated using actual Stardust entry parameters and compared with NASA's Stardust Entry Reconstruction report data to determine code accuracy. All equations used were derived over the course of AE6355.

I. Nomenclature

C_D	=	drag coefficient
C_L	=	lift coefficient
C_A	=	axial force coefficient
C_N	=	normal force coefficient
e	=	eccentricity
D	=	drag force
L	=	lift force
g_o	=	gravitational constant
mw	=	molecular weight
R	=	gas constant
β	=	ballistic coefficient
σ	=	bank angle
k	=	Boltzmann constant or Sutton-Graves constant
ro	=	planet radius
T	=	temperature
P	=	pressure
ρ	=	density
μ	=	gravitational parameter
ω	=	planet's angular rotation
$J2$	=	Jeffery constant
V	=	velocity
t	=	time
M	=	mach number
H	=	scale height
h	=	height
h_{atm}	=	height at atmospheric interface
r_n	=	nose radius
r_c	=	cone base diameter
δ_c	=	sphere cone angle
$\dot{q}''$	=	heat rate
J_{stag}	=	total heat load at the stagnation point
n_{max}	=	peak deceleration

*Graduate Student, Georgia Institute of Technology, Department of Aerospace Engineering

II. Introduction

For any space mission, one of the most critical phases is the entry descent and landing phase (EDL phase). Whether the mission is the return of samples from an unmanned entry vehicle or landing astronauts on mars or Earth, it's the job of EDL engineers to make sure the cargo makes it to the ground in one piece. From determining feasible entry trajectories to TPS material selection and sizing, EDL engineers must be able to adapt to requirements of various mission conditions. The goal of this report is to give an overview of what EDL engineers must consider when supporting entry missions as well as how to analyze potential flight trajectories to find an optimal flight path.

Over the course of the summer term in AE6355, we have discussed the most common methods and trade-offs you can expect to see in current entry, descent and landing missions. Using the many equations derived over the semester, a sample planetary entry simulation code was created through MATLAB to model the multiple trajectories a given spacecraft could take and the resulting forces and heating it would experience. The spacecraft model is based of Stardust using spacecraft geometry from NASA reports. Although there are various entry methods such as steep lifting entry and equilibrium glide, for the purposes of this entry analysis we will focus on a standard direct entry with no skipping. To determine the accuracy of the assumptions used to derive these simplified trajectory analysis equations, the sample simulation will also be compared with Stardust entry reconstruction data.

III. Background on Stardust

Stardust was a robotic space probe launched by NASA on February 7, 1999 and it's primary mission was to collect cometary dust samples from comet Wild 2. Early in 2004 Stardust encountered comet Wild 2 and collected samples of cometary dust and volatiles and began its journey back to Earth. Two years later, the samples made it back to Earth in a return capsule that landed in the Utah desert. The Stardust mission samples indicated that some comets may have included materials ejected from the early sun and may have formed very differently than scientists had previously theorized. The Stardust mission spacecraft is derived from the SpaceProbe deep space bus developed by Lockheed Martin Astronautics.

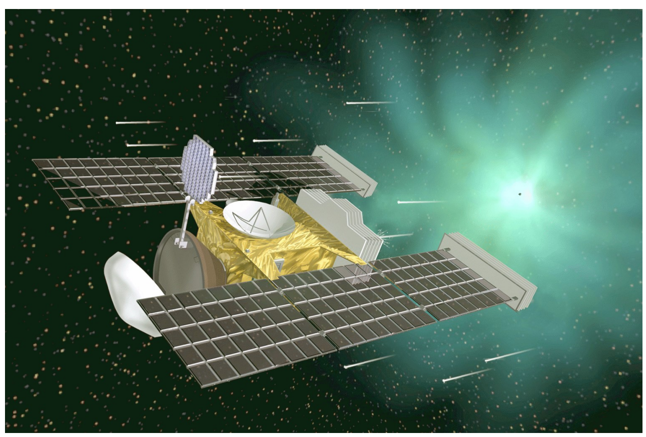

Fig. 1 STARDUST spacecraft model [1].

The total weight of the spacecraft including propellant needed for deep space maneuvers was approximately 380 kilograms. The science payload consisted of aerogel sample collectors, a comet and interstellar dust analyzer (CIDA), a navigation camera (NavCam), and dust flux monitors (DFM). Since Stardust was on a low-energy trajectory for its flyby of comet Wild 2 and return to Earth, Stardust was able to make use of a monoprop propulsion system. The key component of Stardust being analyzed in this paper however, is the sample return capsule. The sample return capsule (SRC) was a 60-degree blunt body reentry capsule. The capsule was encased in PICA and SLA-561 ablator materials to protect the samples in its interior. Near the end of its descent, a parachute was also deployed from the entry capsule to further reduce velocity before touching down. The Stardust entry vehicle had a mass of 46 kilograms, a nose radius of 0.2202 meters, base diameter of 0.8128 meters, and sphere-cone angle of 60 degrees.

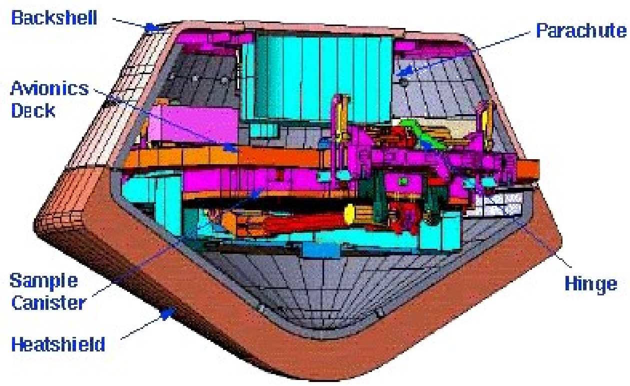

Fig. 2 Sample Return Capsule [1].

IV. Approach

The base mission behind this code is a generic Earth entry mission from any orbit. Using the geometry specified above for the Stardust sample return capsule, the code accepts user inputs for lift over drag, ballistic coefficient, bank angle, max deceleration limit, altitude at periapsis, initial orbit eccentricity, and atmospheric entry velocity with the freedom to specify the flight path angle at atmospheric entry as well. Furthermore, the code is designed to accept inputs to change planet of entry, atmosphere model, and gravitational model. The base test case inputs used for general simulation are provided below in Table 1.

Table 1 Generic Entry Test Conditions

L/D	β	σ	V_{atm}	n_{max}	h_p	e
0.005	120 kg/m^2	0°	8 km/s	30 g's	400 km	0.1

A. Layered Atmosphere Model

The layered atmosphere model used in the code is based off of the 1976 standard atmosphere and is only used with the Earth entry option. The density at a given altitude is determined from the values of temperature and pressure at the bottom of the corresponding layer for which the provided altitude lies within. For altitudes greater than 86 kilometers, the model makes use of a geometric altitude at the bottom of each layer (denoted by subscript i) to determine the density. The equations used to find density above 86 kilometers are provided below [2].

$$\rho = \rho_i * \left[\frac{T_m}{T_{mi}}\right]^{[POW]_\rho} * e^{\left[\frac{g_o b}{RL_{zi}}(z - z_i)\right]} \tag{1}$$

$$[POW]_\rho = -\left[\frac{g_o}{RL_{zi}}\left(\frac{RL_{zi}}{g_o} + 1 + b\left(\frac{T_{mi}}{L_{zi}} - z_i\right)\right)\right] \tag{2}$$

For altitudes less than 86 kilometers, the model makes use of a geopotential altitude at the bottom of each layer to determine the density. Since geopotential is a fictitious altitude, a relationship is needed to connect geometric altitude to geopotential and is provided below in equation 3 (where Γ is equal to $1\frac{m}{m\prime}$). Once related to geometric altitude, the density can be calculated using the ideal gas law [2].

$$h = \Gamma\left(\frac{r_o z}{r_o + z}\right) \tag{3}$$

$$P = \rho RT \tag{4}$$

1. Exponential Atmosphere

As an alternative to the layered atmosphere model to calculate density, the option to use an exponential atmosphere model is also provided. This option allows the user to calculate entry trajectories for both Earth and Mars and assumes temperature (T) is constant throughout the atmosphere. As an alternative, it provides a much simpler way to calculate density at a certain altitude but is less accurate than the layered atmosphere alternative for Earth. The density using the exponential atmosphere model can be calculated as show below in equation 5 where ρ_o is the density at the surface of the planet [2].

$$\rho = \rho_o e^{\frac{-h}{H}} \tag{5}$$

B. Inverse Square Gravity Law

In order to improve the simulation accuracy, the inverse square gravity law was applied to get a better approximation of the gravity. By assuming a spherical gravity field, a gravity value could be calculated for each time step's change in altitude over the course of the decent. The equation of the inverse square gravity law is provided below [2].

$$g = g_o\left(\frac{r_o}{r_o + h}\right)^2 \tag{6}$$

1. J2 Gravity Model

For an even higher fidelity gravity model, the option to use the J2 gravity model is also provided. By accounting for the bulge around the planet's equator, the J2 model provides a better estimate for gravity than the inverse square gravity law. While we can increase our simulation's accuracy by using more than the 1st Jeffrey constant, for simplicity we only consider J2 and neglect longitudinal variations. The J2 constants for both Earth and Mars are provided in Table 2 to model descent trajectories for both planets.

Table 2 J2 Constants

	Earth	Mars
J2	$1.08263 * 10^{-3}$	$1.95545 * 10^{-3}$

The simplified equations of the J2 gravity model are provided below where K is the ellipticity, D is the deviation of the normal, and equatorial radius is expressed as a function of ellipticity and geodetic latitude (ϕ) [2].

$$g = \frac{\mu}{R^2}\left[1 - \frac{3}{4}J_2[1 - 3cos(2\phi)]\right] - R\omega^2 cos(\phi)cos(\phi - D) \tag{7}$$

$$R_E = R_{equator}[1 - Ksin^2(\phi)] \tag{8}$$

$$D = K\left(1 - \frac{h}{R_E}\right)sin(2\phi) \tag{9}$$

C. Planar Equations of Motion

For the actual modeling of the trajectory, the code utilizes the 3 degree of freedom planar equations of motion for constant mass. It's important to note that these equations were solved assuming a negative flight path angle. Since numerical integration is necessary, the code implements the ode45 function in MATLAB to solve iteratively the equations of motion over a specified time interval. The ode45 function takes inputs of time interval specified as a vector from zero to time ti in seconds as well as initial conditions at the start of the decent specified to be at the atmospheric interface. There are three initial conditions necessary in order for the numerical integration to run. Those are, the velocity at atmospheric interface (V_{atm}), the flight path angle at atmospheric entry (γ_e), and the altitude of the atmospheric interface (h_{atm}). The ode45 function is also set to terminate when the numerical integration reaches a negative altitude so as to stop when the simulation reaches the surface of the planet. The equations of motion for negative flight path angle are provided below [2].

$$\frac{dV}{dt} = -\frac{\rho V^2}{2\beta} - gsin\gamma \tag{10}$$

$$\frac{d\gamma}{dt} = \frac{Vcos\gamma}{(ro+h)} + \frac{\rho V}{2\beta}(\frac{L}{D})cos\sigma - \frac{gcos\gamma}{V} \quad (11)$$

$$\frac{dh}{dt} = Vsin\gamma \quad (12)$$

Since the inputs to the numerical integration require a gravity value (g) and density value (ρ) which must be updated at each time step, the above processes for finding gravity and density were inputted in the numerical integration section with ode45.

D. Defining Initial Orbit and Deorbit Trajectories

Since entry flight path angle is determinate on the initial orbit parameters, the code is built to accept inputs of eccentricity (e) and altitude of periapsis (h_p) to define a complete initial orbit that computes the deorbit position and necessary dV to enter the desired decent trajectory. Since it is generally in a mission's best interest to minimize fuel usage, the minimum dV approach was used to define the deorbit trajectories (assume $\gamma_1 = \gamma_2$). Possible departure true anomalies (θ_D) are tested for feasibility based off their corresponding entry flight path angles. The entry flight path angles are bounded by the entry corridor defined by the overshoot and undershoot bounds. The undershoot boundary is defined by the maximum deceleration limit or possibly a maximum heat rate or maximum total heat load a spacecraft can sustain. In terms of this particular simulation, the undershoot is defined by a pre-set deceleration limit generally used for robotic mission of 30 g's. Should the user decide to test possible manned mission entry trajectories, the n_{max} can be adjusted to 10 g's. To determine the steepest entry flight path angle that would still meet the specified deceleration requirement, equation 13 was used. Since the equation is based off of the Allen and Eggers trajectory equations, it is important to note that flight path angle is assumed to be a positive input and the inputs for altitude and velocity are in term of meters [2].

$$n_{max} = \frac{V_{atm}^2 sin\gamma_{atm}}{53.3H} \quad (13)$$

The overshoot boundary is conversely defined by the shallowest possible entry flight path angle that a spacecraft can successfully enter the atmosphere with. Should the flight path angle be too shallow, the spacecraft will deflect off of the planet's atmosphere. In order to find the shallowest entry angle, the 3 degree of freedom planar equations of motion were numerically integrated with a very shallow entry angle as an initial condition which was gradually increase till the first successful atmospheric entry was produced. The iteration through shallow flight path angles was set to terminate if starting altitude was the maximum altitude achieved during the numerical integration.

1. Additional Optimum Deorbit Trajectories

In addition to choosing a departure position that minimizes total dV necessary to enter within the corridor, the code also determines optimum trajectories that minimize peak deceleration, maximum heat rate, and total heat load experienced by the spacecraft during the atmospheric decent phase. This was achieved by storing the values mentioned prior of all possible entry flight path angles that fell within the corridor. Lastly in an attempt to provide a trajectory that minimizes all at once as best a possible, a statistical approach was used. Cycling through each of the stored corridor values mentioned above, each flight path angle's values were tested to see if they lied within ±15% of the mean range of each variable.

E. Aerodynamic Heat Rate and Total Heat Load

As mentioned before, the aerodynamic heat rate and total heat load are necessary parameters to calculate when it comes to heat shield creation. Thus the code calculates the heat rate over the course of the decent at the stagnation point where heating is often at its greatest, denoted $\dot{q}''_{stag}$. The stagnation point heat rate was calculated using the Sutton-Graves correlation since it is usable for any planetary atmosphere and will allow the user to calculate values for both Earth and Mars. The Sutton-Graves constants used for both Earth and Mars are provided below in Table 3. Furthermore, the maximum heat rate along the decent trajectory could easily be determined both graphically or analytically using the Sutton-Graves correlation provided below [2].

$$\dot{q}''_{stag} = k(\frac{\rho}{r_n})^{1/2}V^3 \quad (14)$$

Table 3 Sutton-Graves Constants used for both Earth and Mars

	Earth	Mars
k	$1.74153 * 10^{-4}$	$1.9027 * 10^{-4}$

Once the heat rate over the decent trajectory was calculated, the total heat load was found by simply integrating the heat rate over time using the trapezoid rule. Once all important parameters were calculated, the generic base test case variables specified above were changed one by one to study their effects on the overall mission simulation. This was done by holding all but one variable constant including entry flight path angle and comparing outputs for each trial. The entry flight path angle used in the base test case for these comparisons was that of the flight path angle corresponding to the minimum dV trajectory which was equal to -1.1835 degrees.

1. Aerodynamic Coefficients

Lastly as an additional calculation, the code provides a way to calculate the aerodynamic coefficients as a function of mach number and angle of attack assuming a modified Newtonian flow. For this calculation, the shocks were considered to be normal and properties after the shock were solved for iteratively using the provided equilibrium flow library. The equilibrium flow library was used to solve for the temperature and molecular weight of the air mixture after the shock, and the iteration was set to stop once the solution converged. The equations used to calculate the coefficients of lift and drag after assuming modified Newtonian flow for properties after the shock are provided below [2].

$$C_L = C_N cos\alpha - C_A sin\alpha \tag{15}$$

$$C_D = C_N sin\alpha + C_A cos\alpha \tag{16}$$

V. Results

As discussed above, a series of simulations were run against a base test case to determine the effects of changing certain variables as well as Stardust mission verification test. Starting with the base test case with unspecified entry flight path angle, the entry corridor was found to be between -1 and -10.5 degrees as defined by the overshoot and undershoot bounds shown below in Table 4. All base case calculations were done using the options for Earth, layered atmosphere model, and inverse square gravity law unless specified otherwise.

Table 4 Entry Corridor

	Overshoot	Undershoot
γ_e	$-1.1805°$	$-10.4360°$

In addition to defining the entry corridor bounds, a sample initial orbit was defined using user inputted values for altitude of periapsis and eccentricity. This allowed for a range of potential entry flight path angles to be tested by changing departure true anomalies. As for the minimum dV trajectory, the fully defined initial orbit of the corresponding entry flight path angle was calculated and all notable parameters are provided in the table below (Table 5). The departure point and required dV are also provided.

Table 5 Initial Orbit and Departure Point

r_p (km)	p (km)	r_a (km)	a (km)	ϵ (km^2/s^2)	r_d (km)	θ_D	ΔV (km/s)
6,778	7,455.8	8,284.2	7,531.1	-26.4636	6,868.8	328.7129°	0.3645

Once the entry corridor was defined, optimum trajectories for minimum dV, heat rate, total heat load, and maximum deceleration were plotted on the same graph to determine if it was possible to optimize all parameters at once. Figure 3 shows the results of the simulation plotting both optimum trajectories and the overshoot and undershoot trajectories. As can be seen in Figure 3, minimum heat rate and total heat load trajectories fall on opposite sides of the entry corridor.

This means as you optimize one, the other becomes less optimum. As such, there is a clear trade off in choosing to reduce maximum heat rate or total heat load. On the other hand, Figure 3 would also suggest that it is possible to optimize both heat rate and peak deceleration at the same time as they share very similar entry flight path angles of -1.2133 and -1.1835 degrees respectively. An overall optimum trajectory is also included which occurs at an entry flight path angle of -5.4008 degrees which was found using the statistical approach outlined prior.

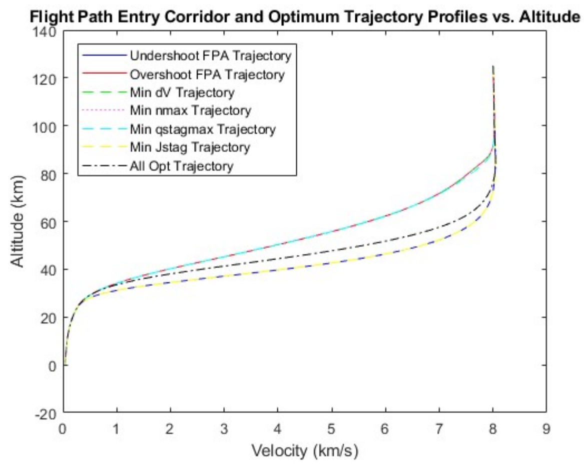

Fig. 3 Possible Optimum Trajectories.

Once all optimum trajectories were plotted, a more detailed analysis of the heat rate and total heat load for the minimum dV trajectory was created by plotting the Sutton-Graves stagnation point heating at each time interval. As can be seen in Figure 4, the peak heat rate occurs about 11 and a half minutes after atmospheric entry reaching a total of about 122 W/cm^2 after which it rapidly drops off. Also provided below in Figure 4 is the total heat load over time which represent the area under the heat rate curve found using the trapezoid rule. As expected, the total heat rate curve is exponential and levels off at about 13 minutes after atmospheric entry reaching a total of about $3.59 * 10^4$ J/cm^2.

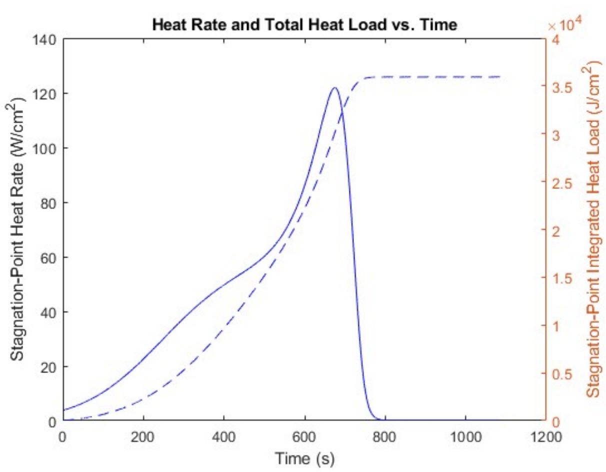

Fig. 4 Min dV Heat Rate and Total Heat Load.

Next, the effects of changing select input values was tested after choosing a fixed entry flight path angle of -1.1835 to serve as a base case input. A table of variables tested is provided below along with their original values (Table 6).

A total of 5 tests were run to make an adjust with only changing one variable at a time while leaving the remaining variables constant. As can be seen in Table 7, when L/D is increased, the maximum heat rate decreases whereas the total heat load increases. For an increase in ballistic coefficient, both maximum heat rate and total heat load can be seen to increase as expected. Next, the data shows that if you were to increase the bank angle, the maximum heat rate would

Table 6 Base Test Parameters

L/D	β	σ	V_{atm}	γ_{atm}
0.005	120 kg/m^2	0°	8 km/s	$-1.1835°$

increase slightly and the total heat rate would decrease slightly. This however might be contradictory as one would expect maximum heat rate to decrease with increasing bank angle and total heat load to increase with the flight duration increase. When atmospheric entry velocity is decreased, the peak deceleration at the set entry angle increases and the departure point radius decreases. In addition, the maximum heat rate increases but the total heat rate decreases. Since both maximum heat rate and total heat load are known to decrease with decreasing atmospheric velocity, this may point to a possible error within the code. Lastly, as entry flight path angle steepens, the max deceleration increases as well as the maximum heat rate. Conversely, the total heat load decreases as well as the departure radius.

Table 7 Simulated Test Cases

Test Cases	n_{max} (g's)	$\dot{q}''_{stagmax}$ (W/cm^2)	J_{stag} (J/cm^2)	r_d (km)	Valid Trajectory (Y/N)
Base	3.4207	121.8558	35,930	6,868.8	Y
L/D=0.05	3.4207	112.3865	38,394	6,868.8	Y
β=200	3.4027	162.3970	50,476	6,868.8	Y
σ=20°	3.4207	121.9385	35,916	6,868.8	Y
V_{atm}=6	23.5108	135.4239	7,269.1	6778	Y
$\gamma_{atm}=-10.4348°$	29.9968	389.9481	10,136	7,937.5	Y

Since the code also provides the option of testing both Mars entry, J2 gravity model, and exponential atmosphere model, a quick simulation was run to show differences between simulation options. First considering Earth, a table has been provided comparing key variables for both atmosphere and gravity models using the base conditions specified above (Table 8). From the data, the majority of values lie within the same range minimal difference. When using the exponential atmosphere model, there is less difference between gravity models and it can noted that the layered J2 model varies the most from the rest. This may likely be due to a small unknown error within the layered atmosphere model.

Table 8 Difference in Selected Models for Earth

Earth	$\dot{q}''_{stagmax}$ (W/cm^2)	J_{stag} (J/cm^2)	n_{max} (g's)
Layered & Inverse	121.8558	35,930	3.4207
Layered & J2	135.3494	26,401	1.3458
Exponential & Inverse	125.9829	30,773	2.8960
Exponential & J2	125.4579	34,554	3.4207

The same was done for Mars excluding the layered atmosphere option assuming that it would not be accurate in modeling the Martian atmosphere. In order to simulate a viable trajectory, the atmospheric velocity and entry flight path angle were changed to be 1.5 km/s and -10.9037 degrees in addition to changing the altitude of periapsis to 200km. The other base conditions were kept the same. The results are provided in Table 9 below. It's important to note that although spacecraft was able to enter the atmosphere, this particular trajectory is not optimal or possibly feasible and was used just for calculation reference.

Table 9 Difference in Selected Models for Mars

Mars	$\dot{q}''_{stagmax}$ (W/cm^2)	J_{stag} (J/cm^2)	n_{max} (g's)
Exponential & Inverse	15.1134	1,061.8	2.0925
Exponential & J2	15.1121	1,062.1	2.0925

As mentioned above, the code also provides a way to calculate the aerodynamic coefficients given an input for mach number and angle of attack. Assuming a modified Newtonian flow iterative approach in combination with the equilibrium flow library, a mach number of 1.37 was selected based on the Stardust parachute deployment mach number as well as an arbitrary angle of attack of 20 degrees. Using the base conditions and the selected values for mach number and angle of attack, the aerodynamic coefficients were calculated and are provided below in Table 10. The result is a noticeably high drag coefficient and low lift coefficient. The L/D assuming modified Newtonian flow was calculated to be -0.1318.

Table 10 Aerodynamic Coefficients

C_N	C_A	C_L	C_D	$\frac{C_L}{C_D}$
0.1541	0.6958	-0.0932	0.7066	-0.1318

A. Verification

Once the general entry simulations were completed, a test using the actual Stardust entry data was run to determine the accuracy of the code. As specified in the Stardust Entry Reconstruction report as well as the entry vehicle reference guide, the entry reconstruction can best be modeled by the following parameters in Table 11 [1][3].

Table 11 Base Test Parameters

L/D	β	σ	V_{atm}	γ_{atm}	$n_{max} limit$
0	60.4 kg/m^2	0°	12.8 km/s	$-8.23°$	32.9 g's

Comparisons for the overall altitude vs velocity plot, altitude and velocity at peak deceleration, magnitude of peak deceleration, peak aerodynamic heat rate, and total heat load are provided below. As can be seen in Figure 5 and 6, the plots of velocity and altitude over time are very similar with a slight difference in initial and final velocities. In addition, Figure 7 and 8 show the plots of altitude versus velocity as well as the stagnation point heat rate and total heat load. Since there were no comparable images for these graphs, the accuracy will be based off peak values.

A table is provided below for easier comparison between Simulated and actual values (Table 12).

Table 12 Comparison of Simulated and Actual Values

	$\dot{q}''_{stagmax}$ (W/cm^2)	J_{stag} (J/cm^2)	n_{max} (g's)	h_{nmax} (km)	V_{nmax} (km/s)
Simulation	785.164	23,067	36.7275	50.2758	8.2397
Actual	817	26,210	32.89	N/A	11.1
% error	3.89	11.99	11.66	N/A	25.76

As can be seen from the comparison, the code is not 100% precise but it does give a decent estimate which is most likely due to the assumption made in the equation used. The equation used to create this are at their most simplified form and thus can lead to a decent amount of error. Nevertheless, they are good to use when in the initial design phase to get a ballpark estimate and that can be shown here in these results.

VI. Conclusion

Over the course of AE6355 we dervived several equation under basic assumption to be able to model EDL behaviors. Although they are strong at getting a rough estimate of many important parameters, there is still room for higher fidelity solutions. As a base code simulation solution, the results were accurate but it would always be best to use higher fidelity simulations in the future.

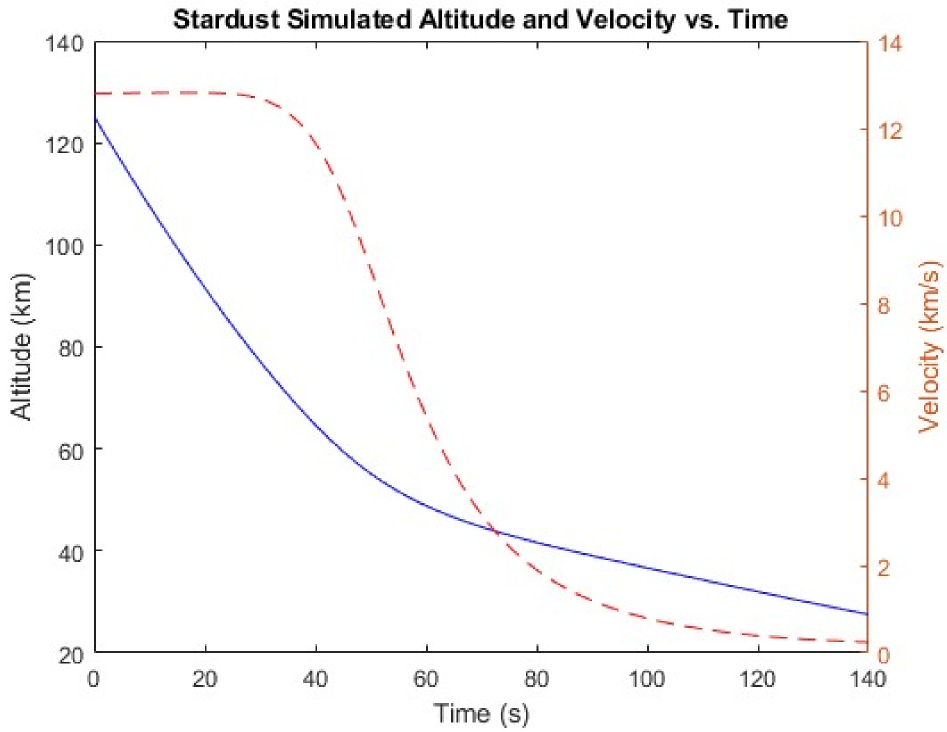

Fig. 5 Simulated Altitude and Velocity over Time.

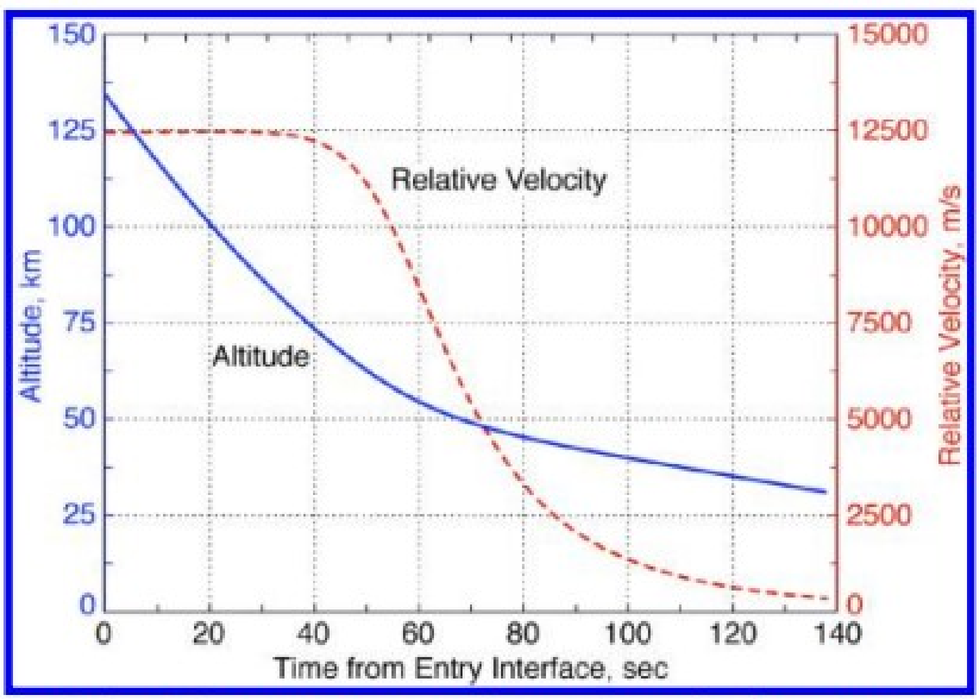

Fig. 8 Altitude and planet-relative velocity profiles from the BET.

Fig. 6 Actual Altitude and Velocity over Time.

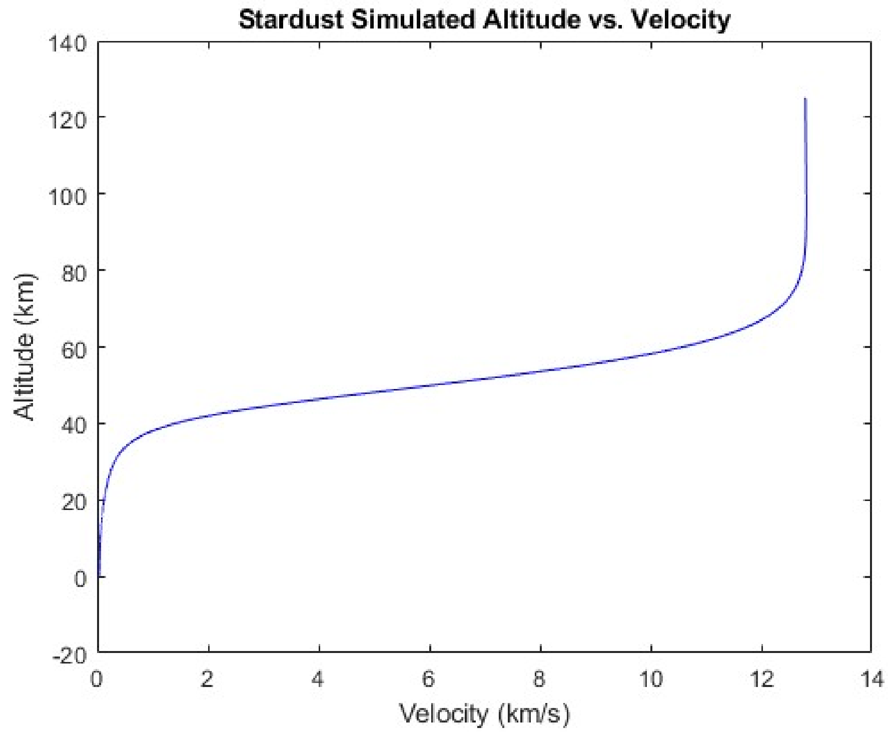

Fig. 7 Simulated Altitude vs Velocity.

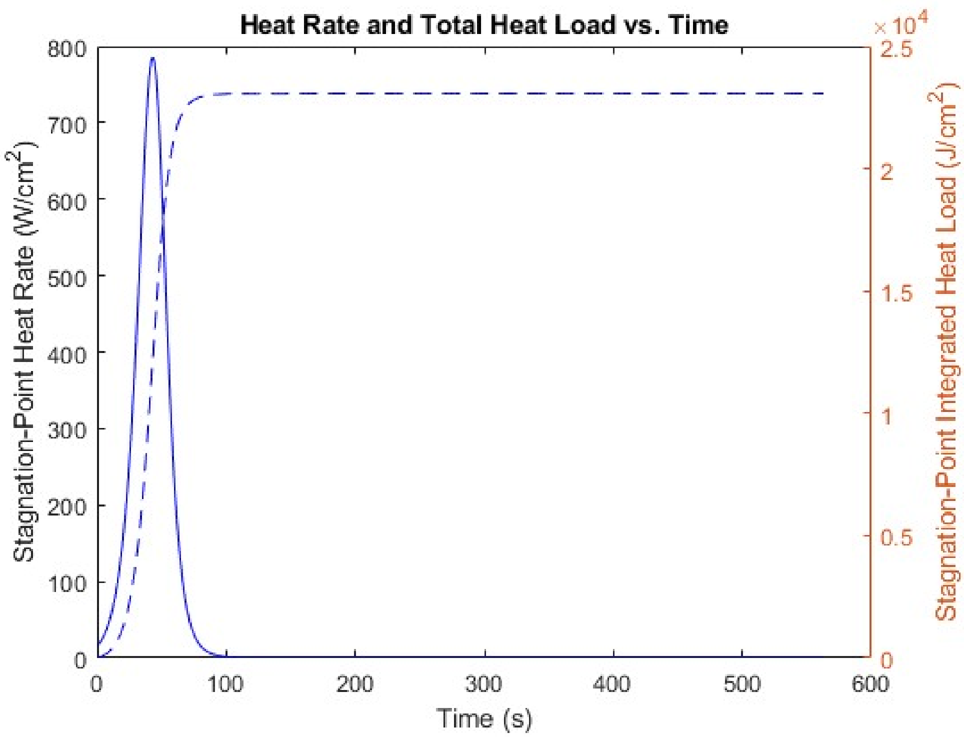

Fig. 8 Simulated Heat Rate and Total Heat Load over Time.

References

[1] NASA-JPL, "STARDUST Mission," , 2003. URL https://solarsystem.nasa.gov/stardust/mission/spacecraft.html.

[2] Dec, J., "AE6355 Class Notes," , 2024. URL https://gtvault-my.sharepoint.com/personal/jdec3_gatech_edu/_layouts/15/Doc.aspx?sourcedoc={62418741-5980-4530-8f5f-fef769d6af4f}&action=view&wd=target%28_Content%20Library%2FUsing%20the%20Content%20Library.one%7C89e5dbea-ffa9-426b-9f19-66f1a9841fca%2FLecture%204.2%20Deorbit%20Example%2C%20RK4%20numerical%20integration%2C%7C14aa720f-713d-4667-b21b-ee19166becfe%2F%29&wdorigin=NavigationUrl.

[3] NASA, "Planetry Entry Vehicles Quick Reference Guide Version 4.0," , 2002. URL https://ntrs.nasa.gov/citations/20220010761.